

LOST IN THE MIDDLE:

HOW LANGUAGE MODELS USE LONG CONTEXTS

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Research Overview

This research examines how language models use their input context.

Objective

The model performance is evaluated on multi-document question answering. To evaluate the relevance of an input context, the length of this context and the relevant informations posititon is changed in a controlled manner.

Data Highlight

Input Context

Write a high-quality answer for the given question using only the provided search results (some of which might be irrelevant).

Document [1] (Title: Asian Americans in science and technology) ...

Document [2] (Title: List of Nobel laureates in Physics) ...

Document [3] (Title: Scientist) ...

Document [4] (Title: Norwegian Americans) ...

Document [5] (Title: Maria Goeppert Mayer) ...

Question: who got the first nobel prize in physics

Answer:

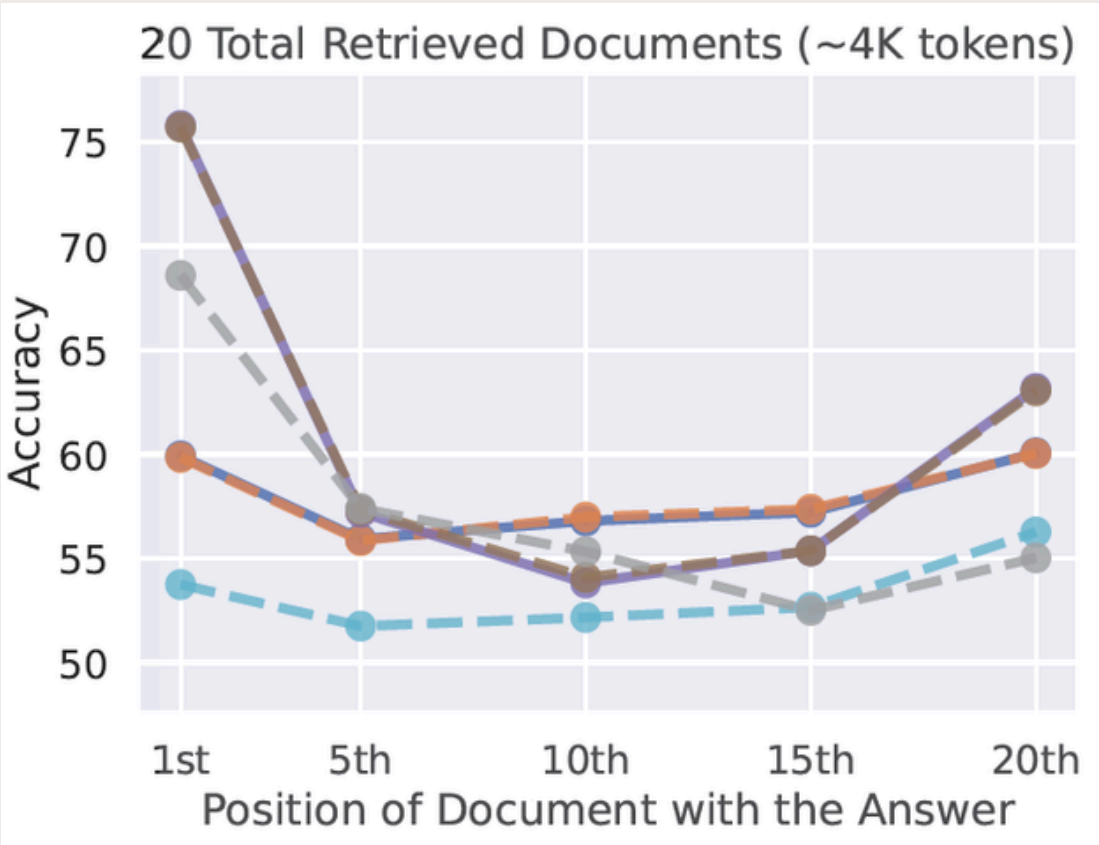
Desired Answer

Wilhelm Conrad Röntgen

Methodology

- Data: NaturalQuestions-Open
- Task: Multi-Document QA
- Variables & Metric: Position, Context Length, Accuracy

Key Insights



Conclusion

- Models perform highest when relevant information is at beginning or end of input context
- Extended-context not necessarily better than input context

FOUND IN THE MIDDLE

MITIGATION STRATEGIES

Context Denoising in Attention

REVISITING LONG-CONTEXT MODELING FROM CONTEXT DENOISING PERSPECTIVE

Objective

- model attention as a signal + noise decomposition where noise grows with context length
- reweight $\text{softmax}(QK^T)$ to enhance high-confidence alignments and dampen diffuse ones (attention sharpening)
 - this reduces entropy of the attention distribution, filtering out low-SNR tokens that dominate in the middle of context regions
- this is applied progressively across layers, so each layer refines and denoises representations
 - results in increasing signal-to-noise ratio (SNR), enabling relevant middle tokens to remain dominant despite heavy context clutter

Multi-Scale Positional Encoding (Ms-PoE)

Found in the Middle: How Language Models Use Long Contexts Better via Plug-and-Play Positional Encoding

Objective

- an effective strategy to downscale positional information
- RoPE encodes position as $f(x, m) = x e^{im\theta}$
 - this position is gets substituted with $m/r \rightarrow$ this forces long-distance tokens to reside in shorted range, allevating the long-term decay effect by factor of r